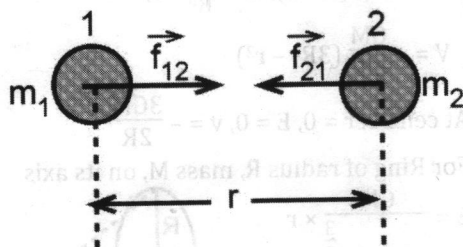


Chapter-11

GRAVITATION

Newton's law of gravitation:



$$|\vec{f}_{12}| = |\vec{f}_{21}| = f(\text{say})$$

$$F \propto \frac{m_1 m_2}{r^2} \Rightarrow F = \frac{km_1 m_2}{r^2}$$

where k is constant known as universal gravitational constant 'G', which only depends on the system of units.

$$F = \frac{G m_1 m_2}{r^2} \quad G = 6.67 \times 10^{-11} \text{ Nm}^2/\text{kg} = 6.67 \times 10^{-8} \text{ dyne cm}^2/\text{g}^2$$

- Gravitation force is always attractive
- Gravitational force is **central force**, **conservative force** acting along the line joining of two bodies.
- If $m_1 = m_2 = 1\text{kg}$, $r = 1\text{m}$ then, $F = G$
 $[G] = \text{M}^{-1} \text{L}^3 \text{T}^{-2}$
- If two identical solid spheres are in contact with each other then force of attraction between them is $F \propto r^4$ where r = radius of sphere
- $g = \frac{GM}{R^2}$ where M = mass of earth

R = Radius of earth

$$g \propto \frac{1}{R^2} \text{ if } M \text{ is constant}$$

$$\% g = -2 \times \% R$$

- If ' ρ ' be the mean density of earth the acceleration due to gravity,

$$g = \frac{4}{3} \pi \rho GR$$

$$g \propto R \Rightarrow \text{i.e. } \% g = \% R$$

Also for inside earth $g \propto r$ where ' r ' is the distance from earth centre

variation of Accel^n due to gravity (g)

with altitude:

g_h = acceleration due to gravity at height ' h '

g = acceleration due to gravity at surface

$$\frac{g_h}{g} = \left(\frac{R}{R+h} \right)^2 \text{ i.e., } g_h < g$$

- If $h \ll R$ then $g_h = g \left(1 - \frac{2h}{R} \right)$
i.e., wt of body decrease with altitude
- If a body is taken from earth surface to height ' h ' so that weight of the body becomes $\frac{1}{n}$ of original wt. then,

$$h = (\sqrt{n} - 1)R$$

- (i) For half wt. $\left(\frac{w}{2} \right)$; $n = 2$, $h = 0.414R = 41.4\%$ of R

$$\text{For } \frac{w}{3}, n = 3, h = 0.732R = 73.2\% \text{ of } R$$

- If w_0 is weight of a body at earth surface then weight of the body at height ' h '.

$$w_h = \frac{w_0}{\left(1 + \frac{h}{R} \right)^2}$$

- (ii) **with depth:**

$g_d = \text{Accel}^n$ due to gravity at depth ' d ' from surface

$g = \text{Accel}^n$ due to gravity at surface

$$\frac{g_d}{g} = \frac{R-d}{R} = \left(1 - \frac{d}{R} \right)$$

$$\text{i.e., } g_d < g \Rightarrow g_d = g \left(1 - \frac{d}{R} \right)$$

- At centre of earth, $d = R$, $g_{\text{centre}} = 0$
- wt. of any body at centre is zero.
- wt. of any body at depth ' d ' $w_d = w_0 \left(1 - \frac{d}{R} \right)$
- If a body is taken from earth surface to depth ' d ' so that weight of the body becomes $\frac{1}{n}$ of original wt. then,

$$d = \frac{(n-1)}{n} R$$

$$\text{For } \frac{w}{2}; n = 2, d = \frac{1}{2} R = 50\% \text{ of } R$$

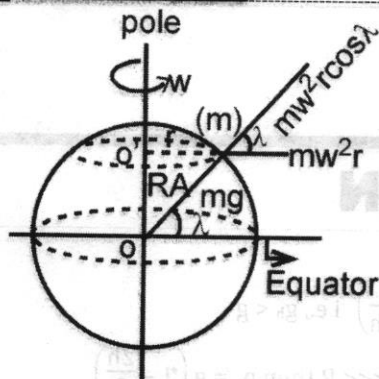
$$\text{For } \frac{w}{3}; n = 3, d = \frac{2}{3} R = 66.67\% \text{ of } R$$

- (iii) **with Rotation:**

λ = latitude

$r = R \cos \lambda$

$$\text{App. wt.} = mg - m\omega^2 r \cos \lambda$$



or, $mg' = mg - m\omega^2 R \cos^2 \lambda$

▪ $g' = g - \omega^2 R \cos^2 \lambda$

▪ At equal $\lambda = 0$, $g_{\text{equator}} = g - \omega^2 R = g_{\text{min}}$

loss in wt = $m\omega^2 R$;

loss in $g = \omega^2 R = 0.034 \text{ m/s}^2$

▪ At pole $\lambda = 90^\circ$, $g_{\text{pole}} = g = g_{\text{max}}$

▪ Due to rotation of earth acceleration due to gravity at all place decreases except pole.

▪ If earth stops its rotation then 'g' at all places increases except pole. At equator by 0.034 m/s^2 .

▪ A body at equator becomes fly off (i.e., weightlessness or apparent wt. zero) if the rotation of earth will be

$$\omega' = \sqrt{\frac{g}{R}} = 17\omega \quad [g' = 0]$$

= 17 times of present value

$$T' = 2\pi \sqrt{\frac{R}{g}} = 84.6 \text{ min} = 1.41 \text{ hr}$$

(iv) Due to shape: Earth is not perfectly spherical.

$$R_{\text{equator}} = R_{\text{pole}} + 21 \text{ km}$$

$$g \propto \frac{1}{R^2}$$

$$g_{\text{equator}} < g_{\text{pole}}$$

♦ Gravitational Field and potential:

▪ Fore experienced by unit mass is the gravitation field

$$E = \frac{F}{m_0} \Rightarrow E = -\frac{dv}{dr}$$

$$\text{potential } V = \int -E \cdot dr$$

(i) For a point mass at a distance 'r'

$$E = \frac{GM}{r^2} \quad V = -\frac{GM}{r}$$

(ii) For Hollow sphere of mass 'M' having radius 'R'

a. out side point ($r > R$), $E = \frac{GM}{r^2}$; $V = -\frac{GM}{r}$

b. on surface ($r = R$), $E = \frac{GM}{R^2}$; $V = -\frac{GM}{R}$

c. for internal point ($r < R$) $E = 0$;

$$V = \text{constant} = -\frac{GM}{R}$$

(iii) For solid sphere of mass M having radius 'R'

a. outside point ($r > R$); $E = \frac{GM}{r^2}$,

$$V = -\frac{GM}{r}$$

b. on surface ($r = R$); $E = \frac{GM}{R^2}$;

$$V = -\frac{GM}{R}$$

c. for inside, ($r < R$); $E = \frac{GM}{R^3} \cdot r$,

$$V = -\frac{GM}{2R^3} (3R^2 - r^2)$$

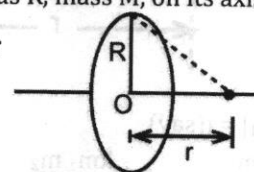
At centre, $r = 0$, $E = 0$, $v = -\frac{3GM}{2R}$

▪ For Ring of radius R, mass M, on its axis

$$E = \frac{GM}{(R^2 + r^2)^{\frac{3}{2}}} \times r$$

At centre, $r = 0$, $E = 0$

$$V = -\frac{GM}{(R^2 + r^2)^{\frac{1}{2}}}$$



▪ If two point masses 'm₁' and 'm₂' are placed at a distance 'r' then gravitation field intensity becomes zero at distance of

a. from m₁;

$$r_1 = \frac{\sqrt{m_1}}{\sqrt{m_1} + \sqrt{m_2}} \times r = \frac{r}{1 + \sqrt{\frac{m_2}{m_1}}}$$

b. from m₂; $r_2 = \frac{r}{1 + \sqrt{\frac{m_1}{m_2}}}$

Motion of satellites:

If a satellite is revolving around the earth in circular path of radius 'r' then centripetal force required by satellite is provided by gravitation force between earth and satellite.

▪ Orbital velocity of satellite (v_0) = $\frac{GM}{r}$

where m = mass of earth $v_0 \propto \frac{1}{\sqrt{r}}$

$$\%v_0 = -\frac{1}{2} \times \%r$$

▪ orbital velocity does not depends on the mass of satellite.

$$\text{Time period, } T = \frac{2\pi}{\sqrt{GM}} \times r^{\frac{3}{2}}$$

▪ If 'h' is the height of satellite from earth surface then,

$$r = R + h$$

For Nearer satellite: The satellite which is very close to earth surface,

$$h \ll R; r = R$$

$$(V_0)_{\text{nearer}} = \sqrt{gR} = 8 \text{ km/sec}$$

$$T = 2\pi \frac{R}{g} = 84.6 \text{ min}$$

- If $F \propto \frac{1}{r^n}$ 'or' r^{-n} then, $T \propto r^{\frac{n+1}{2}}$

Energy of satellite

$$\text{PE, } U = -\frac{GMm}{R} \text{ where } m = \text{mass of satellite}$$

$$\text{KE, } K = \frac{GMm}{2r}$$

$$\text{TE} = K + U = -\frac{GMm}{2r}$$

Here, (-) sign indicates that satellite is bounded to earth. i.e., energy is required to escape the satellite.

$$\text{PE} = 2\text{TE} = -2\text{KE}$$

Binding Energy: Min^m energy required to escape the satellite from the gravitational field.

$$\text{BE} = \frac{GMm}{2r} = \text{KE}$$

Geo-stationary satellite: Such satellite appears stationary w.r.to earth

- (i) Orbit is known as parking orbit
- (ii) Time period, $T = 24 \text{ hr}$
- (iii) Radius of parking orbit, $r = 4.24 \times 10^7 \text{ m}$
- (iv) Height from earth surface, $h = r - R = 3.6 \times 10^7 \text{ m} = 36,000 \text{ km}$

- If is used to transmitting and receiving signals, communication whether forecasting etc.

- If a body is projected from earth surface with velocity V then maximum height attained by body.

$$h = \frac{R}{\left(\frac{2gR}{V^2} - 1\right)}$$

- If $h \rightarrow \infty$, then velocity of projection is known as escape velocity, ($v = v_e$)

$$v_e = \sqrt{2gR} = \sqrt{\frac{2GM}{R}} = \sqrt{2} (v_0)_{\text{nearer}}$$

for earth, $v_e = 11.2 \text{ km/sec}$

for moon, $v_e = 2.4 \text{ km/sec} < \text{RMS velocity of gas}$

(this is the cause of no atmosphere on the moon)

$$v_e = \sqrt{\frac{8\pi \rho G R^2}{3}}$$

- If a body is projected with velocity $v = n v_e$ ($n > 1$) then velocity of the body at infinity (or free space)

$$V_{\text{free space}} \text{ or } V_{\text{infinity}} = (\sqrt{n^2 - 1}) v_e$$

- Max^m height in terms of escape velocity

$$h = \frac{R}{\left(\frac{V_e^2}{V^2} - 1\right)}$$

- If orbital velocity of satellite is increased by 41.4% (or KE by 100%) then satellite escape from its orbit.

(i) If $v < v_0$ then body falls to earth surface in **spiral path** and total energy is (-) ve.

(ii) If $v = v_0$ then body revolves in circular path and $\text{TE} = -\text{ve}$.

(iii) If $v_0 < v < v_e$ body revolves in elliptical path and $\text{TE} = -\text{ve}$

(iv) If $v = v_e$ body escape from gravitation field in parabolic path. $\text{TE} = 0$.

(v) If $v > v_e$ then body will escape in hyperbolic path and $\text{TE} > 0$.

- A body of mass 'm' is at height 'h' from earth surface then increase in potential energy 'or' change in potential energy.

$$\Delta U = \frac{GMm \times h}{R(R+h)} = mgR \frac{h}{R+h}$$

(i) If $h = nR$, then $\Delta U = mgR \times \frac{n}{1+n}$

(ii) If $h \ll R$, then $R+h = R$, $\Delta U = mgh$

Keplar's law for planetary motion

(i) All the planets are revolving around the sun as focus in elliptical path.

(ii) Equal area sweeps out in equal time period.

i.e. A real velocity $\left(\frac{dA}{dt}\right)$ is constant.

(conservation of angular momentum)

$$V \times r = \text{constant}$$

(iii) Square of time period is directly proportional to cube of average distance between plane and sun (i.e., semi major axis)

- Space vehicles are launched from **west to east** in equatorial plane.

- Potential energy required to take the body from r_1 to r_2

$$\Delta U = GMm \left(\frac{1}{r_1} - \frac{1}{r_2} \right)$$

Multiple Choice Questions

1. When escape velocity is given to a particle on surface of earth, its total energy is:

- zero
- greater than zero
- less than zero

- $-\frac{GMm}{2R}$

Gravitation

where M = mass of earth, m = mass of particle and R = radius of earth

2. The ratio of distance of two satellites from the centre of earth is 1: 4. The ratio of their time periods of rotation will be:
 - a. 1: 4
 - b. 4: 1
 - c. 1: 8
 - d. 8: 1
3. for a satellite in elliptical orbit which of the following quantities does not remain constant:
 - a. angular momentum
 - b. momentum
 - c. areal velocity
 - d. total energy
4. An earth satellite is moved from one stable circular orbit to another higher stable circular orbit. Which one of the following quantities increase for the satellite as a result of this change.
 - a. angular momentum
 - b. kinetic energy
 - c. angular velocity
 - d. linear orbital speed
5. Two particles of mass m and M are initially at rest and infinitely separated from each other. Due to mutual interactions they approach each other. Their relative velocity of approach at a separations d between them is:
 - a. $\left[\frac{2Gd}{M+m}\right]^{1/2}$
 - b. $\left[\frac{2G(M+m)}{d}\right]^{1/2}$
 - c. $\frac{2GM+m}{d}$
 - d. none of these
6. the orbital velocity of an artificial satellite in a circular orbit just above the earth's surface is v . For a satellite orbiting at an altitude of half the earth's radius the orbital velocity is:
 - a. $\frac{3}{2}v$
 - b. $\sqrt{\frac{3}{2}}v$
 - c. $\sqrt{\frac{2}{3}}v$
 - d. $\frac{2}{3}v$
7. The escape velocity of a body projected vertically upward from the earth's surface is 11.2 km/s. If the body is projected in a direction making 30° angle to the vertical, its escape velocity in this case will be:
 - a. 11.2 km/s
 - b. $\frac{11.2}{2}$ km/s
 - c. $11.2 \times \frac{\sqrt{3}}{2}$ km/s
 - d. $\frac{11.2}{3}$ km/s

8. Inside a uniform spherical shell:

- a. potential energy is zero
- b. field is not defined
- c. potential is constant
- d. all of the above

9. Earth orbiting satellite will escape if:

- a. Its speed is increased by 41%
- b. Its speed in the orbit is made $\sqrt{1.5}$ times of its value
- c. Its KE is doubled
- d. It slopes moving in the orbit

10. An artificial satellite moving in circular orbit around the earth has a total (kinetic + potential) energy E_0 . Its PE is:

- a. $-E$
- b. $1.5 E_0$
- c. $2E_0$
- d. E_0

11. A satellite in an equatorial orbit has a time period of 6 h. At a certain instant, it is directly overhead an observer on the equator of the earth. It is directly overhead the observer again after a time T . The possible value of T is:

- a. 12 h
- b. 8 h
- c. 1'8 h
- d. 36 h

12. A simple pendulum has a time period T_1 when on the earth's surface, and T_2 when taken to a height R above the earth's surface, where R is the radius of the earth. The value of $\frac{T_2}{T_1}$ is:

- a. 1
- b. $\sqrt{2}$
- c. 4
- d. 2

13. at what height the gravitational field reduces by 75% the gravitational field at the surface of earth?

- a. R
 - b. $2R$
 - c. $3R$
 - d. $4R$
- (where, R is the radius of earth)

14. two bodies of masses m and M are placed a distance d apart. The gravitational potential at the position where the gravitational field due to them is zero is V .

- a. $V = -\frac{G}{d}(m+M)$
- b. $V = -\frac{Gm}{d}$
- c. $V = -\frac{GM}{d}$
- d. $V = -\frac{G}{d}(\sqrt{m} + \sqrt{M})^2$

15. The gravitational potential energy of a body at a distance r from the centre of earth is u . Its weight as a distance $2r$ from the centre of earth is:

- a. $\frac{u}{r}$ b. $\frac{u}{4r}$
c. $\frac{u}{2r}$ d. $\frac{4r}{u}$

16. Two satellites of same mass are launched in the same orbit around the earth so as to rotate opposite to each other. If they collide elastically and stick together as wreckage, the total energy of the system just after collision is:

- a. $-\frac{2GMm}{r}$ b. $-\frac{GMm}{r}$
c. $\frac{GMm}{2r}$ d. $\frac{GMm}{4r}$

17. The weights of an object in a coal mine, at sea level, and at the top of mountains are w_1 , w_2 and w_3 respectively. Then:

- a. $w_1 < w_2 > w_3$ b. $w_1 = w_2 = w_3$
c. $w_1 < w_2 < w_3$ d. $w_1 > w_2 > w_3$

18. A thin rod of length L is bent to form a semicircle. The mass of rod is M . What will be the gravitational potential at the centre of the circle?

- a. $-\frac{GM}{L}$ b. $-\frac{GM}{2\pi L}$
c. $-\frac{\pi GM}{2L}$ d. $-\frac{\pi GM}{L}$

19. If V is the gravitational potential on the surface of the earth then what is its value at the centre of the earth?

- a. $2V$ b. $3V$
c. $\frac{3}{2}V$ d. $\frac{2}{3}V$

20. The rotation of the earth having radius R about its axis speeds its axis speeds up to a value such that a man at latitude angle 60° feels weightlessness. The duration of the day in such a case is:

- a. $\pi\sqrt{\frac{R}{g}}$ b. $4\pi\sqrt{\frac{R}{g}}$
c. $2\pi\sqrt{\frac{g}{R}}$ d. $4\pi\sqrt{\frac{g}{R}}$

21. A body has a weight 72 N. When it is taken to a height $h = R =$ radius of earth, it would weigh:

- a. 72 N b. 36 N
c. 18 N d. zero

22. A body of mass m is placed on earth surface. It is then taken from earth's surface to a height $h = 3R$, then the change in gravitational potential energy is:

- a. $\frac{mgh}{R}$ b. $\frac{2}{3}mgR$
c. $\frac{3}{4}mgR$ d. $\frac{mgR}{2}$

23. The acceleration due to gravity on planet A is 9 times the acceleration due to gravity on planet B. A man jumps to a height of 2m on the surface of A. What is the height of jump by the same persons on the planet B?

- a. 6 m b. $\frac{2}{3}$ m
c. $\frac{2}{9}$ m (d) 18 m

24. An artificial satellite is revolving around the earth in a circular orbit. If its speed is half of the escape velocity. Then its height above the surface of earth of radius R , is:

- a. $\frac{R}{4}$ b. $\frac{R}{2}$
c. $2R$ d. R

25. A planet of mass m moves around the sun of mass M in an elliptical orbit. The maximum and minimum distances of the planet from the sun are r_1 and r_2 respectively. The time period of the planet is proportional to:

- a. $r_1^{2/5}$ b. $(r_1 + r_2)^{3/2}$
c. $(r_1 - r_2)^{3/2}$ d. $r^{3/2}$

26. A satellite is launched into a circular orbit of radius R around the earth. While a second satellite launched into an orbit of radius $1.01R$. The period of the second satellite is longer than the first one by approximately:

- a. 3.0 % b. 1.5 %
c. 0.7 % d. 1.0 %

Gravitation

27.If the radius of earth's orbit is made one fourth, the duration of a year will become:

- a. 1/4 times b. 1/8 times
c. 4 times d. 8 times

28.An earth satellite S has an orbit radius which is 4 times that of communication satellite C. The period revolution of S will be.

- a. 32 days b. 18 days
c. 8 days d. 9 days

29.If the potential energy of a body on a planet is numerically U and the escape velocity for the same body be v_e for the same planet then $\frac{U}{v_e}$ will be :

- a. $\frac{U}{v_e} = \sqrt{\frac{GM}{2R}}$
b. $\frac{U}{v_e} = m \sqrt{\frac{GM}{2R}}$
c. $\frac{U}{v_e} = m \sqrt{\frac{2GM}{R}}$
d. $\frac{U}{v_e} = m \frac{GM}{R}$

30.If the change in the value of g at a height h above the surface of the earth is the same as at a depth x below its surface, then:

- a. $x = h^2$ b. $x = 0.4h$
c. $x = 2h$ d. $x = h$

31. Suppose a tunnel is dug along the diameter of the earth. A particle is dropped from a point directly above the tunnel. If the earth's density is assumed to be uniform and the friction is neglected, then choose the wrong answer:

- a. Particle will have maximum speed when passing through the centre of the earth
b. Acceleration of particle will be the maximum just at the point of release
c. Particle will have simple harmonic oscillation.
d. Particle will drop to the centre of the earth

32.Four equal masses (each of mass M) are placed at the corners of a square of side a. The escape velocity of a body is from the centre of the square is:

- a. $4\sqrt{\frac{2GM}{a}}$ b. $\sqrt{\frac{8\sqrt{2}GM}{a}}$
c. $\frac{4GM}{a}$ d. $\sqrt{\frac{4\sqrt{2}GM}{a}}$

33.The escape velocity from earth is v_0 . If the mass of a certain planet is 3 times and radius 3 times that of earth, then the escape velocity from the planet will be :

- a. $3v_0$ b. $6v_0$
c. $\sqrt{3}v_0$ d. v_0

34.A satellite of mass m, moving around the earth in a circular orbit of radius R, has angular momentum L. The areal velocity of satellite is: (M_e = mass of earth)

- a. $L/2m$ b. L/m
c. $2L/m$ d. $2L/M_e$

35.If the earth loses its gravity then for a body:

- a. weight becomes zero but not the mass
b. mass becomes zero but not the weight
c. both mass and weight becomes zero
d. neither mass no weight becomes zero.

36.The escape velocity of a body from the earth is 11.2 km/sec. If the radius of planet be half the radius of earth and its mass be one fourth that of the earth. The escape velocity of the planet will be:

- a. 2 km/sec b. 16 km/sec
c. 4 km/sec d. 8 km/sec

Answers:

1. a	2. c	3. b	4. a	5. b
6. b	7. a	8. c	9. a	10. c
11. b	12. d	13. a	14. d	15. b
16. a	17. a	18. d	19. c	20. a
21. c	22. c	23. d	24. c	25. b
26. b	27. b	28. c	29. b	30. c
31. d	32. b	33. d	34. a	35. a
36. d				

Solution for Selected Problems

2. $T \propto r^{3/2}$

$$\frac{T_1}{T_2} = \left(\frac{r_1}{r_2}\right)^{3/2} = \left(\frac{1}{4}\right)^{3/2} = 1:8$$

5. By using concept of reduced mass,

Gravitation

$$\therefore T_2 = 8 \text{ days}$$

$$29. -\frac{GMm}{R} + \frac{1}{2}mv_e^2 = 0$$

$$v_e = \sqrt{\frac{2GM}{R}}$$

$$U = \frac{GMm}{R}$$

$$\frac{U}{v_e} = m \sqrt{\frac{GM}{2R}}$$

$$30. \text{At height } g_h = g(1 - \frac{2h}{R})$$

$$\text{At depth, } g_x = g(1 - \frac{x}{R})$$

$$g_h = g_n$$

$$\therefore x = 2h$$

$$32. \text{From conservation of energy,}$$

$$-\frac{GMm}{(a/\sqrt{2})} \times 4 + \frac{1}{2}mv_e^2 = 0$$

$$v_e = \sqrt{8\sqrt{2} \frac{GM}{a}}$$

$$33. V_e = \sqrt{\frac{2GM}{R}}$$

$$V_p = \sqrt{\frac{2GM'}{R'}}$$

$$M' = 3M; R' = 3R$$

$$\therefore V_p = V_e = V_o$$

$$36. V_e = \sqrt{\frac{2GM}{R}}$$

$$V_p = \sqrt{\frac{2GM'}{R'}}$$

$$M' = \frac{M}{4}$$

$$R' = \frac{R}{2}$$

$$V_p = \sqrt{\frac{2GM}{2R}} = \frac{11.2}{\sqrt{2}} = 8 \text{ km/s}$$



$$\mu = \frac{mM}{M+m}$$

$$-\frac{GmM}{d} + \frac{1}{2}\mu v_r^2 = 0$$

$$v_r^2 = \sqrt{\frac{2GmM}{dr}} = \left[\frac{2G(M+m)}{d} \right]^{1/2}$$

$$6. V_0 = \sqrt{\frac{GM}{(R+R/2)}} = \sqrt{\frac{2}{3} \frac{GM}{R}} = \sqrt{\frac{2}{3}} V$$

$$\therefore \text{where, } v = \sqrt{\frac{GM}{R}}$$

11. If satellite and earth rotates in same directions

$$\frac{1}{T} = \frac{1}{6} - \frac{1}{24} = \frac{4-1}{24} = \frac{1}{8}$$

$$\therefore T = 8 \text{ hr}$$

$$12. g_h = g \left(\frac{R}{R+R} \right)^2 = \frac{g}{4}$$

$$T \propto \frac{1}{\sqrt{g}}$$

$$\frac{T_2}{T_1} = \sqrt{\frac{g}{g/4}} = 2$$

$$\therefore \frac{T_2}{T_1} = 2$$

$$13. E = \frac{GM}{R^2}$$

$$\therefore E' = 0.25 \frac{GM}{R^2} = \frac{GM}{4R^2}$$

$$E' = \frac{GM}{r^2} \Rightarrow r = 2R; R+h = 2R$$

$$\therefore h = R$$

$$14. \frac{Gm}{x^2} = \frac{GM}{(d-x)^2}$$

$$\therefore \frac{d-x}{x} = \sqrt{\frac{M}{m}}$$

$$\frac{d}{x} - 1 = \sqrt{\frac{M}{m}}$$

$$\Rightarrow x = \frac{\sqrt{m} d}{\sqrt{M} + \sqrt{m}}$$

$$\text{Potential at point P, } V_p = -\frac{Gm}{x} - \frac{GM}{(d-x)}$$

$$V_p = -\frac{G}{d} (\sqrt{M} + \sqrt{m})^2$$

$$15. U = -\frac{GMm}{r}$$

$$\text{wt. at a distance 'r' from centre, } w = \frac{GMm}{r^2}$$

$$\text{wt. at a distance '2r' from centre, } w' = \frac{GMm}{(2r)^2} = \frac{GMm}{4r^2}$$

$$w' = -\frac{U}{4r} = \frac{U}{4r}$$

16. Since they are moving in opposite direction,

$$mv - mv = (m+m)v' \Rightarrow v' = 0$$

$$KE = 0$$

$$PE \quad U = \frac{-GM(m+m)}{r} = -\frac{2GMm}{r}$$

$$TE = KE + PE = -\frac{2GMm}{r}$$

$$18. \pi R = L \Rightarrow R = \frac{L}{\pi}$$

$$\text{Potential at centre: } V = -\frac{GM}{R} = -\frac{\pi GM}{L}$$

$$19. V = -\frac{GM}{2R^3} (3R^2 - r^2)$$

$$\text{At centre } r = 0, V_c = -\frac{3GM}{2R} = \frac{3}{2} V [V_{\text{surface}} = -\frac{GM}{R}]$$

$$20. g' = g - \omega'^2 R \cos^2$$

$$\Rightarrow 0 = g - \omega'^2 R \cos^2 60^\circ$$

$$\therefore \omega' = \sqrt{\frac{4g}{R}} = 2\sqrt{\frac{g}{R}}$$

$$T' = \frac{2\pi}{\omega'} = \pi \sqrt{\frac{R}{g}}$$

$$21. \text{weight at height 'h': } W_h = \frac{w_0}{\left(1 + \frac{h}{R}\right)^2} = \frac{72}{\left(1 + \frac{R}{R}\right)^2} = 18 \text{ N}$$

$$22. \text{Change in PE; } \Delta U = \frac{mgh}{1 + \frac{h}{R}} = \frac{mg \times 3R}{1 + \frac{3R}{R}} = \frac{3}{4} mgR$$

$$23. g_B \times h_B = g_A \times h_A$$

$$h_B = \frac{9 \times g_B \times 2}{g_B} = 18 \text{ m}$$

$$26. T \propto r^{3/2}$$

$$\begin{aligned} \frac{\Delta T}{T} &= \left(\frac{T_2}{T_1} - 1 \right) \times 100 \\ &= \left[\left(\frac{1+0.01}{1} \right)^{3/2} - 1 \right] \times 100 \\ &= \frac{3}{2} \times 0.01 \times 100 = 1.5\% \end{aligned}$$

$$28. \left(\frac{T_2}{T_1} \right) = \left(\frac{r_2}{r_1} \right)^{3/2}$$

$$\frac{T_2}{1 \text{ day}} = \left(\frac{4r}{r} \right)^{3/2}$$